NETWORKING

Computer Networks — Complete Study Notes

Network Types · Topologies · Networking Devices · OSI Model · IP Addressing · TCP/IP · Data Communication · Transmission Media · RAID & Backup

Typed from the handwritten class notes — every topic, example and diagram retained, with clarifications where the page was unclear or needs a correction.

Contents

1. Computer Networks and their Types 4

1.1 LAN — Local Area Network . . . 4
1.2 CAN — Campus Area Network . . . 4
1.3 MAN — Metropolitan Area Network . . . 4
1.4 WAN — Wide Area Network . . . 4
1.5 PAN — Personal Area Network . . . 5
1.6 SAN — Storage Area Network . . . 5
1.7 BAN — Body Area Network . . . 6
1.8 HAN — Home Area Network . . . 6
1.9 VPN — Virtual Private Network . . . 6
1.10 P2P — Peer-to-Peer network . . . 6
Comparison of network types (from the notes) . . . 6

2. Network Topology 7

2.1 Mesh topology . . . 7
2.2 Bus topology . . . 7
2.3 Star topology . . . 8
2.4 Ring topology . . . 8
2.5 Tree topology . . . 9
Summary of topologies . . . 10

3. Networking Devices 11

3.1 Modem . . . 11
3.2 Repeater . . . 11
3.3 Hub . . . 11
3.4 Switch . . . 11
3.5 NIC — Network Interface Card . . . 12
3.6 Router . . . 12
3.7 Bridge . . . 12
3.8 Gateway . . . 12
3.9 Brouter . . . 12

4. OSI Model 13

Data unit at each layer (the “data flow” table) . . . 13
Layer 1 — Physical layer . . . 13
Layer 2 — Data link layer . . . 14
Layer 3 — Network layer . . . 15
Layer 4 — Transport layer . . . 16
Layer 5 — Session layer . . . 17
Layer 6 — Presentation layer . . . 17

Layer 7 — Application layer . . . 18

5. IP Addressing **19**

5.1 Management of IP addresses . . . 19

5.2 Types of IP address . . . 19

5.3 Versions of IP . . . 19

Summary of IPv4 classes (from the notes) . . . 21

5.4 IP address configuration . . . 22

5.5 IPv6 . . . 22

6. TCP/IP Model **23**

7. Data Communication **24**

7.1 Components of data communication . . . 24

7.2 Data transmission modes (responsibility of the physical layer) . . . 24

7.3 Data communication media . . . 24

8. Backup and Restoring Data **27**

8.1 (A) RAID configuration . . . 27

RAID summary . . . 29

8.2 (B) Backup . . . 29

Appendix — Clarifications and corrections **29**

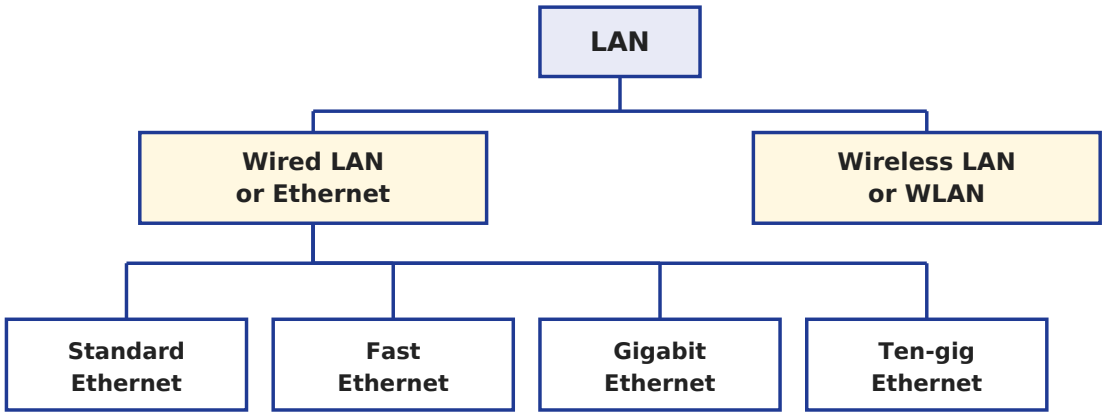
1. Computer Networks and their Types

- A **network** is a medium through which a connection is established between multiple computers — **wired or wireless**.
- Types: (1) LAN, (2) CAN, (3) MAN, (4) WAN, (5) PAN, (6) SAN ... etc.

1.1 LAN — Local Area Network

- **Private and secure**; range up to **2 km**.
- Small geographical area, like a school or office.
- All resources are connected to a central device (**hub or switch**).
- A **router** is used to connect the LAN to the Internet.
- To create a wireless LAN, a **WAP** (Wireless Access Point) is used.

Evolution of LAN



LAN → wired LAN (Ethernet: Standard, Fast, Gigabit, Ten-gig) and wireless LAN (WLAN).

Types of LAN — ARCNET

- **ARCNET** = Attached Resources Computer Network.
- Oldest and easy installation.
- Started in **1977** by Datapoint Corporation, New York.
- Available for microcomputers from **1980**; speed **2.5 Mbps**.
- Uses the **token-passing** mechanism.

1.2 CAN — Campus Area Network

- Range **1 to 5 km**.
- LAN < CAN < MAN.
- A connection of inter-connected LANs.

1.3 MAN — Metropolitan Area Network

- Big geographical area, like a city.
- Works on **IEEE 802.6**.
- Wired MAN uses **optical fibre** (works on 802.8).
- Uses a **data-link-layer** protocol.

1.4 WAN — Wide Area Network

- The **largest network** of the world (known as the **Internet**).
- **Least secure** and low speed.

- Network to network.
- Uses “VLPS (variable length packet switching)” for communication — works on the network layer.
- Uses fibre-optic cable.
- Uses **satellite**:

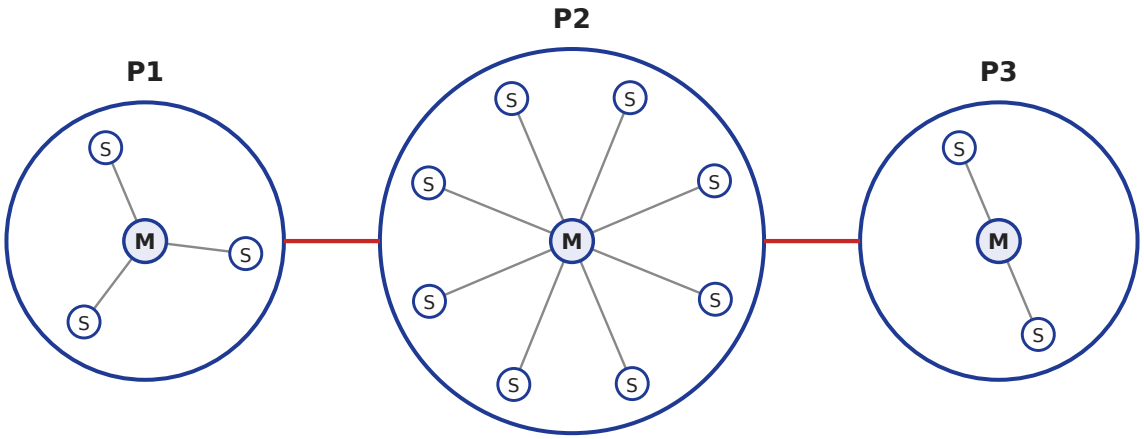
Satellite term	Meaning
Repeaters	Transponders
Uplink	Earth → satellite
Downlink	Satellite → Earth

Clarification

The notes write “VLPS”; the usual name for this WAN technique is **MPLS** (Multi-Protocol Label Switching), which also appears in the network-layer protocol list later in these notes.

1.5 PAN — Personal Area Network

- Known as **WPAN**; works on **IEEE 802.15**.
- **Bluetooth** connectivity — IEEE 802.15.1; range about **10 m**; started by **Jaap Haartsen**.
- **Ad-hoc** network; **7 devices** connect — 1 master device and 6 slave devices.
- Only the master can send data; master and slave can connect with each other.
- Standard transmission capacity about **720 Kbps** (as read from the notes).
- Works on the **2.4 GHz** frequency; low cost and short range.
- The basic architecture of Bluetooth is called a **PICONET**; a group of piconets is a **SCATTERNET**.
- Master-to-master connection is needed to connect multiple piconets; a slave can be used, which acts as a **bridge**.



SCATTERNET (3 M, 13 S)

Piconets P1, P2, P3 joined into a scatternet (3 masters, 13 slaves, as labelled in the notes).

1.6 SAN — Storage Area Network

- Known as **cloud services**; started about **1999** (by Salesforce — SaaS).

Company	Cloud service
Amazon	AWS
Apple	iCloud
Microsoft	Azure
Google	GCP

Cloud service models

Model	Meaning	Points	Examples
IaaS	Infrastructure as a Service	Maximum flexibility; high availability; cost and time saving	Azure, AWS, GCP
PaaS	Platform as a Service	Cost-effective development; high availability; easy migration to the hybrid model	Docker, Kubernetes
SaaS	Software as a Service	Accessibility; cost efficiency; all-inclusive	HubSpot, Office 365, Slack

1.7 BAN — Body Area Network

- A bug / chip is put on a person's body in order to **trace** it.

1.8 HAN — Home Area Network

- A Wi-Fi modem connects home / office devices.

1.9 VPN — Virtual Private Network

- To extend a **private and secure** network over a public network.
- Used to work from home.
- Encrypted connection; blocked websites become reachable; data secured.
- It changes your IP.

1.10 P2P — Peer-to-Peer network

- **Decentralized** network (because there is no server).
- Transfers bigger data easily.
- Each computer works as **client and server**.
- Started in **1979**.

Advantages	Disadvantages
No need of any OS (as written) If any computer fails, there is no problem in the network	No control server No data back-up No security Slow network

Comparison of network types (from the notes)

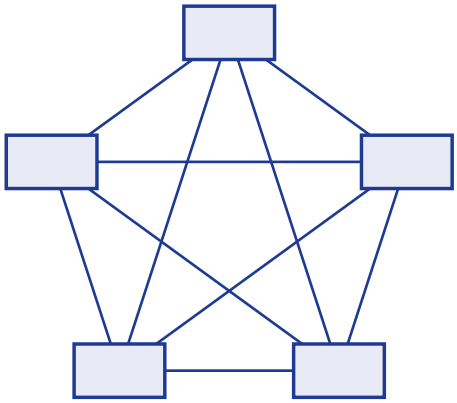
Type	Range / scope	Notes
LAN	up to 2 km — school, office	Private & secure; hub/switch; router to Internet
CAN	1 to 5 km	Inter-connected LANs; LAN < CAN < MAN
MAN	City	IEEE 802.6; optical fibre; data-link layer
WAN	World (Internet)	Least secure, low speed; fibre / satellite
PAN	≈ 10 m	Bluetooth, IEEE 802.15; piconet / scatternet
SAN	Cloud	AWS, iCloud, Azure, GCP
BAN	On the body	Tracing chip
HAN	Home	Wi-Fi modem
VPN	Over public network	Encrypted, secure

2. Network Topology

- Network topology is based on **connection**; two types:
- **(a) Physical topology** — works on the **physical layer** of the OSI model.
- **(b) Logical topology** — works on the **network layer** of the OSI model (as written in the notes).

2.1 Mesh topology

- Every node is connected to every other node.
- Data flow is **bi-directional**.
- Used in **WAN**.
- Fully connected topology; the **most reliable and secure**.
- Decentralized network.
- Nodes are connected with a channel / link / cable.
- **Redundancy and fault tolerance**; multiple paths between two nodes.



5 nodes: cables = 5×4/2 = 10

Formula	Meaning
cables = n(n−1)/2	total cables for n nodes
connections = n − 1	links at each node
I/O ports = 2 × cables	total ports

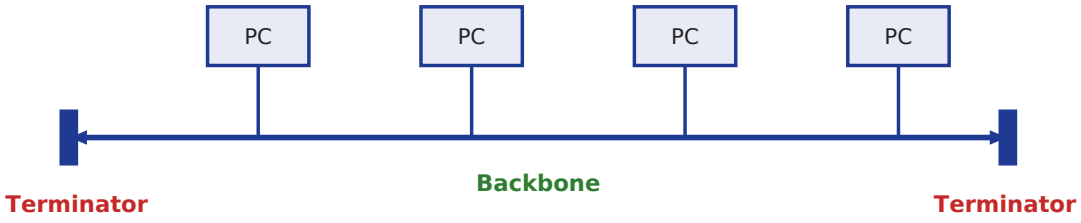
Types of mesh topology

Type	Meaning
Fully mesh	Nodes are fully connected — (n − 1) connections at every node
Partial mesh	Nodes are not fully connected — optional paths are used

Advantages	Disadvantages
One device fails → no full-network failure No traffic problem High privacy and security Data transmission is more consistent — a failed node / cable does not interrupt it Adding a new device does not disturb transmission	Bulk cabling required Installation and configuration are tough Most expensive Each node needs extra utility cost More maintenance required Higher power requirement

2.2 Bus topology

- Connected with a **single cable**; known as **bus / backbone / segment**.
- Line / linear / horizontal; data flow **bi-directional**.
- **Max 30** nodes can connect.
- A **terminator** (vampire tap, as written) is connected at both ends to block data loss.
- **Ethernet** (wired LAN) uses it.

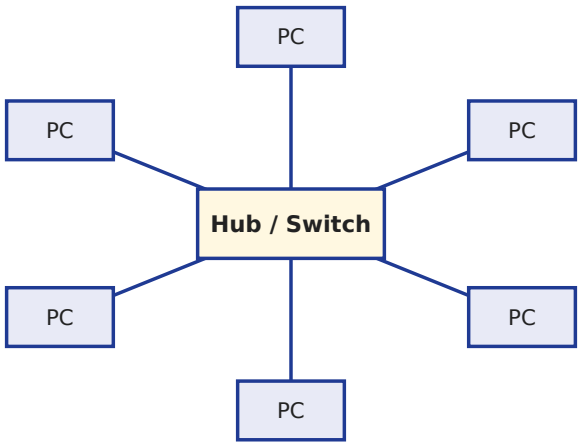


Bus topology: nodes hang off a single backbone cable with a terminator at each end.

Advantages	Disadvantages
Simple and inexpensive Scalable — adding a new device is easy Total cable required = **n + 1**	If the backbone fails, the whole network fails Limited length of network Data collision

2.3 Star topology

- Data **broadcast**; a **hub / switch** is the central device that shares connections point-to-point.
- Hub works on the OSI **physical layer**.
- Data flow — unidirectional.
- Cables = number of nodes.



Star topology: every node connects to the central hub / switch.

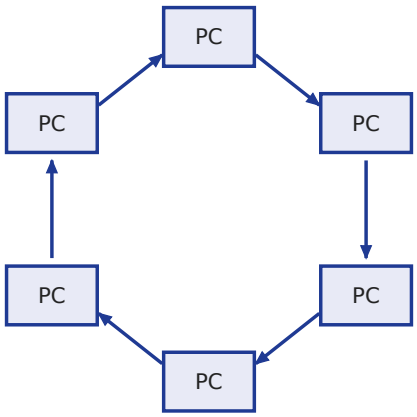
Advantages	Disadvantages
Adding a new node is easy No effect on the network if one node fails Easy to identify a failed node	Central device (server) fails → whole network fails

Type	Details
(a) Broadcast star network	A **hub** is used as the centralised device
(b) Switched star network	A **switch** is used; a modern switch supports both unicast and multicast

2.4 Ring topology

- Each node is connected to the next and the previous node, forming a ring.
- Data passes through **every node**.
- If any single node fails, the whole network fails.
- Data flow **uni-directional**.
- **Token-passing** method is used for data transfer; data transmission is bit-wise.

- Each node has a **repeater** that amplifies the data received from the previous node and then sends it to the next node.
- No end — circular; **RJ-45 or co-axial** cable is used; no need of any centralized device.



Ring topology: data circulates in one direction through every node.

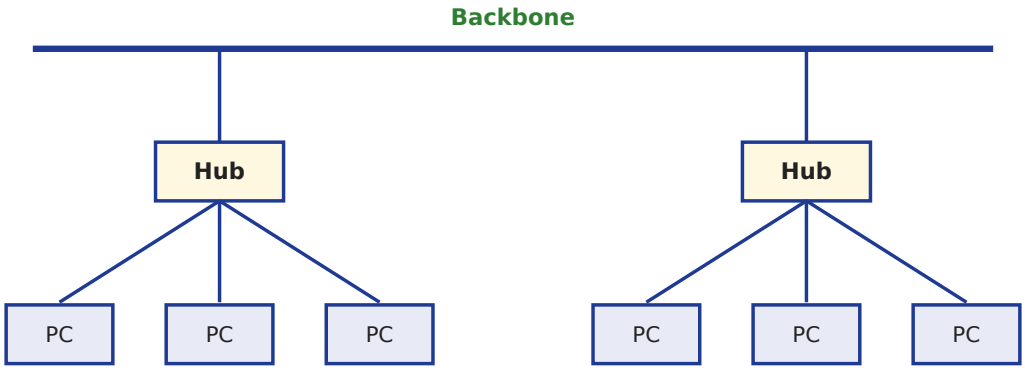
Advantages	Disadvantages
Faster speed, reliable, easy to expand No collision, because of unidirectional flow	Slower than star topology More nodes → less speed One node fails → whole network fails

FDDI — Fibre Distributed Data Interface

- Bi-directional; contains **two rings — primary and secondary**.
- Primary carries data; secondary carries **backup data**.
- Max length = **200 km**; max nodes = **1000**.

2.5 Tree topology

- Combination of **bus and star** topology; **hierarchy** topology.
- Used for WAN; extended version of the star topology.
- If the backbone fails, not all of the network fails.
- Secure topology; best performance than LAN (as written).



Tree topology: hubs (stars) attached to a backbone (bus).

Advantages	Disadvantages
Error detection and correction are easy Secure and reliable	Setup cost high Configuration tough More cable use

Summary of topologies

Topology	Cable / link count	Data flow	Central device	Fails when…
Mesh	n(n−1)/2 cables	Bi-directional	None	Never fully (redundant paths)
Bus	n + 1	Bi-directional	None (backbone)	Backbone fails
Star	n	Uni-directional	Hub / switch	Central device fails
Ring	—	Uni-directional	None	Any single node fails
Tree	—	—	Hubs on a backbone	Backbone fails: only part fails

3. Networking Devices

- Networking devices are the devices used to **connect clients and servers**.

OSI layer	Devices working there
Application layer	Gateway, Firewall
Presentation layer	Gateway
Session layer	Gateway
Transport layer	Gateway, Packet-filter firewall
Network layer	Gateway, Router, Brouter (Bridge + Router), 3-layer switch
Data link layer	Gateway, Bridges, NIC, 2-layer switch
Physical layer	Gateway, Modem, Hub, Repeater, Cables

3.1 Modem

- **Modulator–Demodulator**.
- **Analog → digital = demodulator**; **digital → analog = modulator**.
- Used in star topology; works on the physical layer; telephone lines and computers use it.
- When 2 modems connect to each other it is called **handshaking**.
- Speed is measured in **BPS** (bits per second).
- Also called a **hybrid device** — works as both an input and an output device.

3.2 Repeater

- **Amplifies and regenerates** weak signals; works on the physical layer.
- Used to extend the length of a LAN, or to connect two separate LANs.
- **Non-intelligent**, pure / dumb hardware: only forwarding, no data filter.
- A hub is called a **multiport repeater**.

3.3 Hub

- Works on the **physical layer**; a multiport repeater; non-intelligent device.
- From one node, data is **broadcast** to all connected nodes.

Type	Details
(a) Passive hub	Acts like a connector only; works below the physical layer; does not regenerate signals. (The notes also say “can’t work without electricity”.)
(b) Active hub	Works as a repeater; the most commonly used basic hub in any domain.
(c) Intelligent hub	A type of active hub, but it regenerates and monitors signals; uses **SNMP**; also known as a switch.

Correction to the handwritten notes

A passive hub needs **no** external power (it only joins cables); it is the active and intelligent hubs that need electricity.

3.4 Switch

- **Intelligent device** (has knowledge of MAC addresses); data is transferred to the **specific receiver**.
- Works on the data link layer; 5–52 ports.
- To forward data to the receiver, a switch keeps a table in **CAM** (Content Addressable Memory).

Type	Works on
(a) 2-layer switch — also called a multiport bridge	Data link layer of the OSI model; **MAC address**
(b) 3-layer switch — called a router	Network layer; **IP address**

3.5 NIC — Network Interface Card

- A hybrid device.
- Known as: **physical address, MAC address, hybrid address, burned-in address, chassis no., data-link address**.
- **6 bytes = 48 bits = 12 hexadecimal digits**, written as six pairs with **5 separators** (:).

format: __ : __ : __ : __ : __ : __ (6 pairs of hex digits = 6 bytes, separated by 5 colons), e.g. B6:A6:C2:B5:A2:B5

3.6 Router

- Connects **LAN to WAN**; works on the physical, data link and network layers.
- Creates a **routing table** for all possible receivers; works as a **traffic director** for data packets.
- Uses the **IP address**; connects two different topologies.
- Uses **RIP** (Routing Information Protocol), **OSPF** (Open Shortest Path First) and **BGP** (Border Gateway Protocol).
- Uses a firewall against unauthorised access.

3.7 Bridge

- Works on the **data link layer** of the OSI model.
- Used to **extend a LAN**; can do **data filtration**.
- Uses **MAC addresses** to connect LANs.

3.8 Gateway

- Works through the **whole OSI model and TCP/IP model**.
- Also known as a **firewall** (as written).
- Connects all LANs of different configuration; works as an **entry and exit point**; a **protocol converter**.
- A router works **within** a network, but a gateway lets networks of **different models** work together.

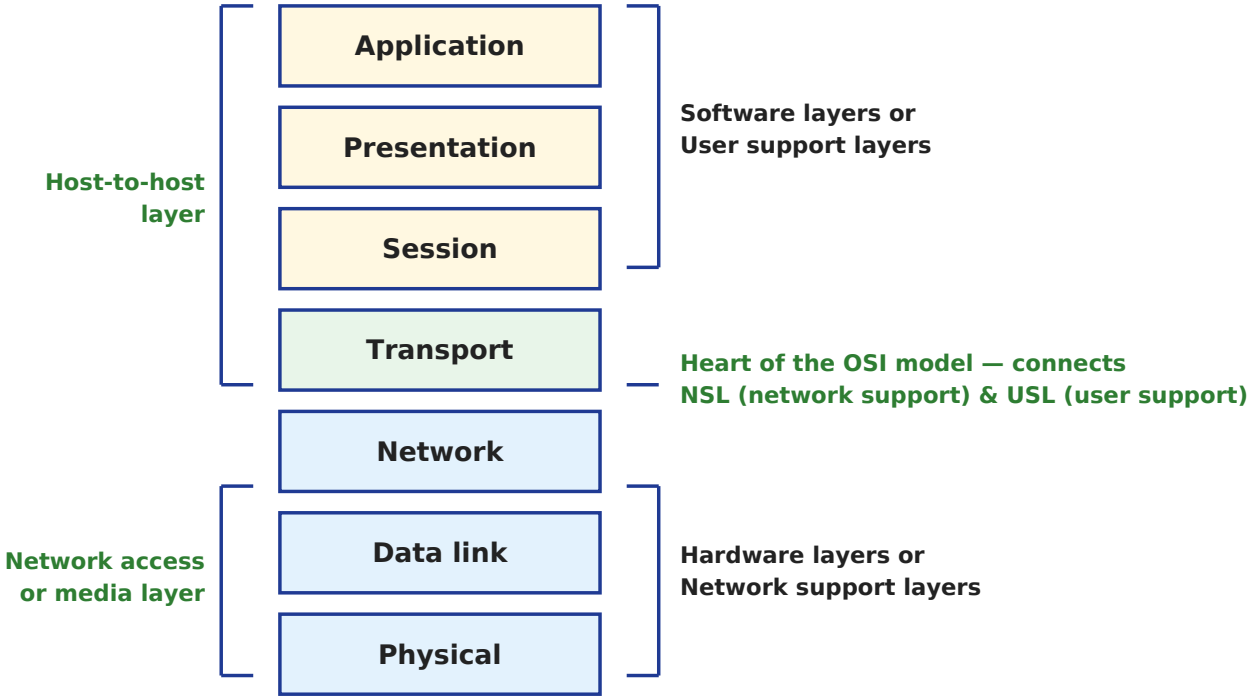
3.9 Brouter

- **Brouter = Bridge + Router**.

Device	Layer	Address used	Intelligent?
Modem	Physical	—	Hybrid I/O device
Repeater	Physical	—	No (dumb)
Hub	Physical	— (broadcast)	No (intelligent hub = switch)
Switch (2-layer)	Data link	MAC	Yes
Bridge	Data link	MAC	Filters data
NIC	Data link	MAC (burned in)	—
Switch (3-layer) / Router	Network	IP	Yes (routing table)
Gateway	All layers	—	Protocol converter

4. OSI Model

- **Open System Interconnection**; also known as the **referential, theoretical, logical, conceptual** model.
- Proposed in **1970** by Charles Bachman; published by **ISO (1984)**.
- Memory trick (bottom → top): **“Please Do Not Take Sabzi, Pooja A…”** — P-D-N-T-S-P-A = Physical, Data link, Network, Transport, Session, Presentation, Application.



The seven OSI layers and how they group (software / hardware layers; transport is the heart of the model).

Data unit at each layer (the “data flow” table)

Layer	Name	Data unit
7	Application	DATA / MESSAGE
6	Presentation	DATA
5	Session	DATA
4	Transport	Segment / Datagram
3	Network	Packets
2	Data link	Frames
1	Physical	BITS

Layer 1 — Physical layer

- The 1st layer — **data bits travel**.

Responsibility

- Defines the **physical topology**.
- Defines data **transmission modes** (simplex, half-duplex, full-duplex).
- Defines the **transmission media**.
- Synchronizes bits to balance source and destination.

Limitations

- No error handling; no data interpretation.
- Susceptible to physical damage.

Protocols / devices

- **ISDN** (Integrated Services Digital Network)
- **SONET** (written “Synchronous Digital Network”; the standard name is Synchronous Optical Network)
- Bluetooth, Hub, Gateway, Repeaters, Modem, Wi-Fi, Infrared
- Ethernet, USB

Layer 2 — Data link layer

- Layer 2 of the OSI model; the data unit is the **frame**.
- Hop-to-hop / node-to-node / point-to-point / link-to-link delivery.

Responsibilities

- Creates the **Ethernet frame**: **max length = 1518 bytes**, **min length = 64 bytes**, **7 fields** in a frame.
- **Error control and flow control**.
- Defines **access control**.
- Uses **MAC addresses** (unicast, multicast, broadcast).

Question — which address is unicast?

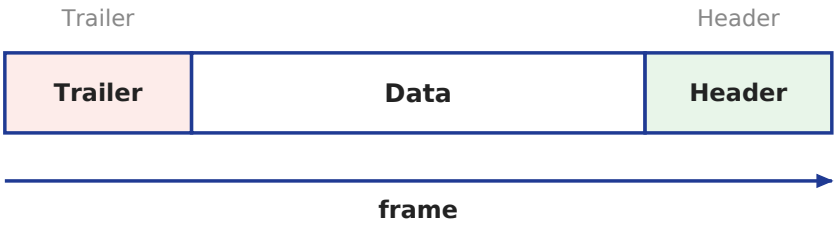
Option	MAC address	2nd digit of 1st byte	Type
(A)	B6:A6:C2:B5:A2:B5	6 (even)	**UNICAST ✔**
(B)	A5:B6:A2:B5:C2:B5	5 (odd)	Multicast
(C)	E3:B5:A2:B6:A5:B5	3 (odd)	Multicast
(D)	C5:A2:B6:B5:A2:B5	5 (odd)	Multicast

Rules

If the **second digit of the first byte** of a MAC address is an **even** number (0 included) the address is **UNICAST**.
If it is **odd** → **MULTICAST**.
If every digit is F → **BROADCAST**, i.e. **FF:FF:FF:FF:FF:FF**.

Error control in layer 2

- Extra bits are added to frames to control errors; it also helps stop duplication of frames (that line of the notes is hard to read).



Frame as drawn in the notes: Trailer at the left end, Header at the right end, with the whole frame below.

Note

In the usual layout a frame is written Header → Data → Trailer from left to right. The notes label the ends the other way round (the arrow “frame” runs left → right under the box), so the figure above follows the notes as drawn.

Limitations of layer 2

- Limited scale.
- Increased overhead.
- Error-handling dependency (for complex errors).
- No routing capabilities.

Protocols

Protocol	Full form
ARP	Address Resolution Protocol
DTP	Dynamic Trunking Protocol
LACP	Link Aggregation Control Protocol
PPP	Point-to-Point Protocol
HDLC	High-level Data Link Control protocol
SDLC	Synchronous Data Link Control protocol
SLIP	Serial Line Interface Protocol
LCP	Link Control Protocol
NCP	Network Control Protocol
Others	Ethernet, FDDI, 2-layer switch, Token Ring, Bridge, NIC, WAP

Layer 3 — Network layer

- Data travels in the form of a **packet**; **source to destination / host to host** delivery; chooses the **best path** to deliver data.

Responsibilities

- **Packet forwarding and routing**.
- Determines the quality and priority of packets being transmitted.
- Provides a unique **IP address** to devices for identification and communication.
- Sends packets to the destination using routers.
- Divides a large data packet into smaller **fragments** for transmission and reassembles them at the receiver’s end.
- Divides a larger network into smaller sub-networks (**subnetting**).

Limitations

- No flow control.
- Routers may lose packets.

Protocols

Protocol	Full form
ICMP	Internet Control Message Protocol
EGP	Exterior Gateway Protocol
IGMP	Internet Group Management Protocol
IGRP	Interior Gateway Routing Protocol
RIP	Routing Information Protocol
VRRP (written VRGP)	Virtual Router Redundancy Protocol
IPv4 & IPv6	Internet Protocol versions 4 and 6
RARP	Reverse Address Resolution Protocol
OSPF	Open Shortest Path First
BGP	Border Gateway Protocol
IPSec	Internet Protocol Security
MPLS	Multi-Protocol Label Switching

Layer 4 — Transport layer

- Layer 4 of the OSI model; data travels as **segments / datagrams**.
- The **heart of the OSI model**; links the NSL (network support layers) and USL (user support layers).
- Reassembles the data at the receiver's end.
- Delivers data **end to end / process to process / machine to machine**.

Responsibility

- Connection control between source and destination.
- **Segmentation and reassembling**.
- Error control and flow control.
- **Multiplexing and demultiplexing**.
- Adds the receiver **port number** in the data header (**service-point addressing**).
- TCP and UDP; end-to-end connection control between hosts.
- **Congestion control**; **data integrity**.

TCP vs UDP

	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Connection	Connection-oriented	Connectionless
Reliability	Reliable & secure	Unreliable & unsecure
Speed	Slow	Fast
Broadcasting	Does not support	Supports broadcasting
Orientation	Byte-oriented	Message-oriented
Weight	Heavy-weight	Light-weight

Protocols

Protocol	Full form
TCP, UDP	—
FCP	Fibre Channel Protocol
DCCP	Datagram Congestion Control Protocol
SCTP	Stream Control Transmission Protocol
ATP	AppleTalk Transaction Protocol
BGP	Border Gateway Protocol
SPX	Sequenced Packet Exchange

Layer 5 — Session layer

- Layer 5; responsible for the **connection / session**.

Responsibilities

- Creation of the connection, management and **termination**.
- **Token management**.
- Data transmission mode.
- **Synchronization** is implemented by inserting **checkpoints** for recovery at the time of disruption.
- Ensures orderly communication; **dialogue control**.

Protocols

Protocol	Full form
NCP	Network Control Protocol
RPC (written “RP (Remote Control)”)	Remote Procedure Call
SIP	Session Initiation Protocol
PPTP	Point-to-Point Tunneling Protocol
ZIP	Zone Information Protocol
NFS	Network File System protocol
PAP	Password Authentication Protocol
RTCP	Real-time Control Protocol

Layer 6 — Presentation layer

- Layer 6; **changes data to a standard format**; concerned with the data **syntax and semantics**.

Responsibilities

- Data **encryption and decryption**.
- Data **compression and decompression**.
- Data **translation** at the destination.

Protocols

Protocol	Full form
SSL	Secure Socket Layer
JPEG	Joint Photographic Experts Group
MPEG	Moving Picture Experts Group
TLS	Transport Layer Security
LPP	Lightweight Presentation Protocol

Layer 7 — Application layer

- Layer 7; **nearest to the user**; the desktop layer.

Responsibilities

- **Email**-related services.
- **Remote login** service.
- **File transfer and manage** (FTAM).

Protocols

Protocol	Full form
AMQP	Advanced Message Queuing Protocol
BOOTP	Bootstrap Protocol
DNS	Domain Name System
DHCP	Dynamic Host Configuration Protocol
FTP	File Transfer Protocol
HTTP, HTTPS	Hypertext Transfer Protocol (Secure)
IMAP	Internet Message Access Protocol
MIME	Multipurpose Internet Mail Extension
NTP	Network Time Protocol
POP-3	Post Office Protocol, version 3
Telnet	Telecommunication Network

5. IP Addressing

- **IP** = Internet Protocol; a **logical address**; works on the **network layer**.
- IP address = **Network ID + Host ID**.

5.1 Management of IP addresses

- IP addresses are managed by **IANA** (Internet Assigned Numbers Authority).
- IANA provides IP addresses and port numbers.
- Administration is done by **RIRs** (Regional Internet Registries) — 5 RIRs:

#	RIR	Full name	Started	Covers
1	**APNIC**	Asia Pacific Network Information Centre	1993	South, East & South-East Asia (India also)
2	**AFRINIC**	African Network Information Centre	2004	All of the African continent
3	**ARIN**	American Registry for Internet Numbers	1997	USA, Canada, Antarctica & Caribbean
4	**LACNIC**	Latin America and Caribbean Network Information Centre	1993	Latin America & Caribbean
5	**RIPE NCC**	Réseaux IP Européens Network Coordination Centre	1992	Europe, Central Asia, Western Asia

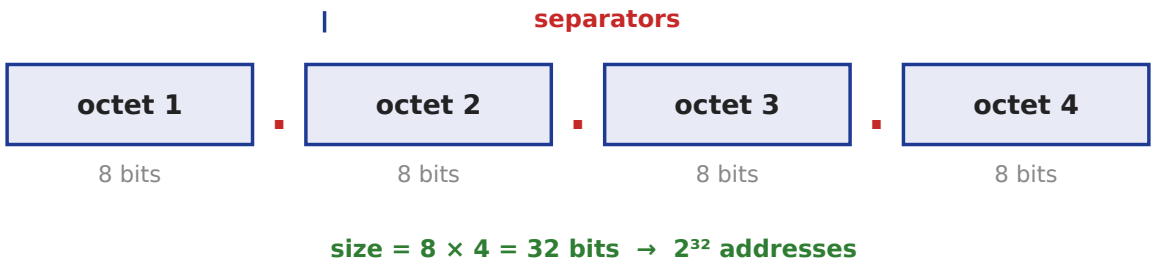
5.2 Types of IP address

#	Type	Details
1	**Public IP**	Provided by the ISP to the router; for direct Internet access
2	**Private IP**	Provided by networking devices like a router
3	**Static IP**	Static — does not change; fixed IP; assigned by the DHCP server
4	**Dynamic IP**	Assigned by the ISP; dynamic IP via DHCP and PPPoE (Point-to-Point over Ethernet)

5.3 Versions of IP

(A) IPv4

- Internet Protocol version 4; created in **1981**.
- Four octets separated by dots (separators).
- Size = **8 × 4 = 32 bits**; maximum addresses = $\mathbf{2^{32}}$.
- Supports unicast, multicast and broadcast.



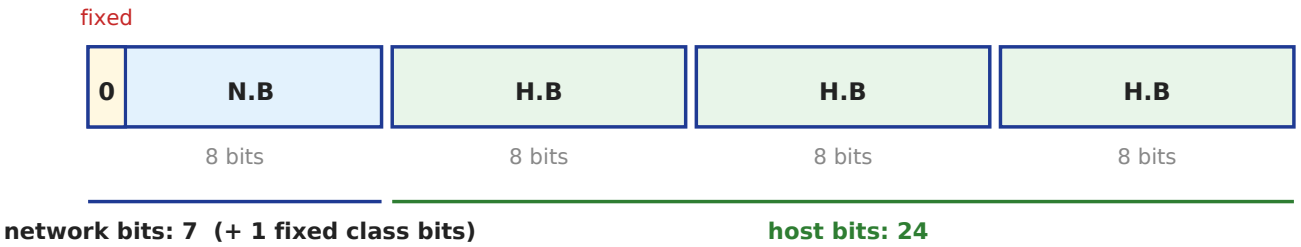
IPv4 address format: four octets of 8 bits joined by separators.

Address classes

(1) Class A

- Designed for the **maximum number of hosts**.
- The **first bit of the first octet is 0**; the remaining 7 bits form the network ID.

- Using the first octet as the network ID, a total of **126 networks** can be created.
- Range = 0 to 127.



Class A: 1 fixed bit (0) + 7 network bits, then 24 host bits.

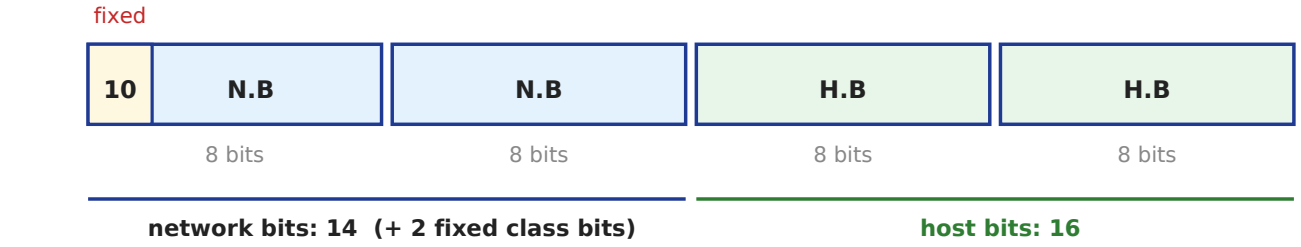


First octet: 00000000 (0) to 01111111 (127) — the first bit is fixed at 0.

- **127 is reserved** for the local loop(back), so the range changes to **0–126** (as written).
- Public IP range = 1.0.0.0 to 127.0.0.0 (as written).
- Private IP range = **10.0.0.0 to 10.255.255.255**.
- Total networks = $2^n - 2 = 128 - 2 =$ **126** (n = network bits).
- Total hosts = $2^h - 2 = 2^{24} - 2 =$ **16,777,214** (h = host bits).
- To get the **subnet mask**: make all network bits **1** and host bits **0** → Class A subnet mask = **255.0.0.0**.
- Used for unicasting.

(2) Class B

- Medium to large-scale networks; the **first 2 octets are network bits**, the last 2 octets are host bits.
- The first 2 bits of the network bits are fixed as **1 and 0**.
- Range = **128 to 191**.



Class B: fixed bits 10 + 14 network bits, then 16 host bits.



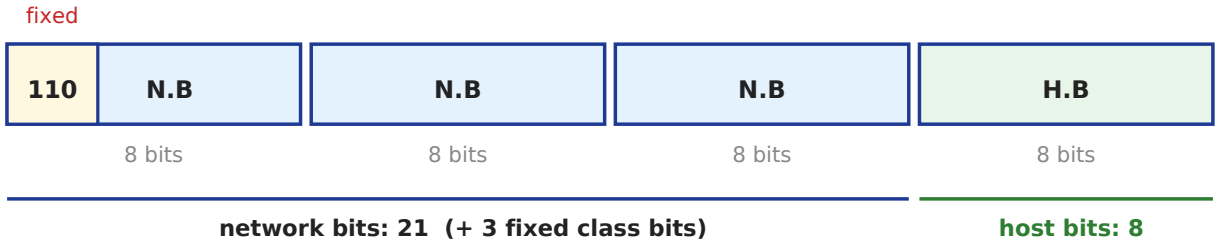
First octet: 10000000 (128) to 10111111 (191).

- Public IP range = 128.0.0.0 to 191.255.0.0.
- Private IP range = **172.16.0.0 to 172.31.255.255**.
- Subnet mask = **255.255.0.0**; unicast.
- Total networks = $2^n - 2 = 2^{14} - 2 = 16384$ (as written; the arithmetic gives 16,382).
- Total hosts = $2^h - 2 = 2^{16} - 2 =$ **65,534**.

(3) Class C

- LAN or small geographical use; the **first 3 octets are network bits**, the last octet is host bits.
- The first 3 bits of the network bits are fixed as **1, 1, 0**.

- Range = **192 – 223**.



Class C: fixed bits 110 + 21 network bits, then 8 host bits.



First octet: 11000000 (192) to 11011111 (223).

- Public IP range = 192.0.0.0 – 223.255.255.0.
- Private IP range = **192.168.0.0 – 192.168.255.255**.
- Subnet mask = **255.255.255.0**; unicast use.
- Total networks = $2^n = 2^{21}$.
- Total hosts = $2^h - 2 = 2^8 - 2$ = **254 hosts**.

(4) Class D

- **Only network bits, no host bits**.
- 4 fixed bits: **1, 1, 1, 0**.
- Range = **224 to 239**; public IP range = 224.0.0.0 – 239.255.255.255.

(5) Class E

- **Not in use**; reserved for research purposes.
- Reserved bits written as **1, 1, 1, 1, 0** (5 bits); range = **240 – 255**.

Clarification

For class E the standard fixed prefix is **1111** (4 bits), which is what makes the range 240–255; a 5-bit prefix 11110 would only give 240–247. Class A's usable range is normally written 1–126 (0 is also special).

Summary of IPv4 classes (from the notes)

Class	Fixed bits	First octet	Network bits	Host bits	Networks	Hosts	Default mask
A	0	0 – 127 (0–126)	7	24	126	16,777,214	255.0.0.0
B	10	128 – 191	14	16	16384*	65,534	255.255.0.0
C	110	192 – 223	21	8	2^{21}	254	255.255.255.0
D	1110	224 – 239	—	—	—	—	—
E	1111(0)	240 – 255	—	—	—	—	—

** as written in the notes ($2^{14} - 2 = 16{,}382$).*

Class	Public IP range	Private IP range
A	1.0.0.0 – 127.0.0.0 (as written)	10.0.0.0 – 10.255.255.255
B	128.0.0.0 – 191.255.0.0	172.16.0.0 – 172.31.255.255
C	192.0.0.0 – 223.255.255.0	192.168.0.0 – 192.168.255.255
D	224.0.0.0 – 239.255.255.255	—

5.4 IP address configuration

- **(A) Automatically (B) Manually**
- Protocols:

(1) DHCP — Dynamic Host Configuration Protocol

- This protocol uses the **UDP** services of the transport layer.
- Server port number — **67**; client port number — **68**.

(2) WINS — Windows Internet Name Service

- When the DHCP server is unavailable, the OS uses an important feature called **APIPA** (Automatic Private IP Addressing).
- APIPA range = **169.254.0.1 to 169.254.255.254**.

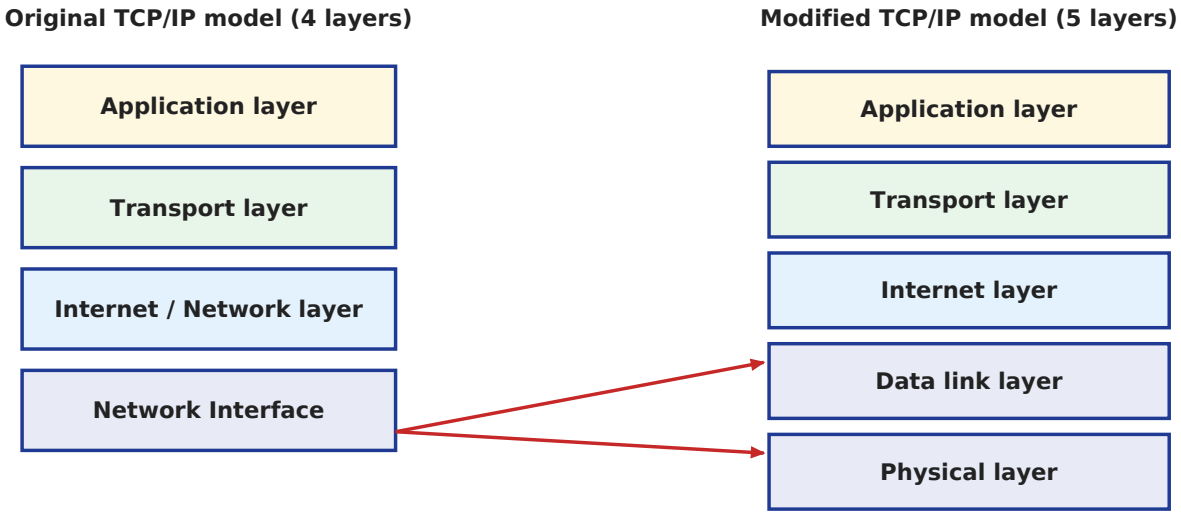
5.5 IPv6

- Created in **1995** by the **IETF** (Internet Engineering Task Force).
- Uses 8 groups (“octets”) of 16 bits: **8 × 16 = 128 bits**, written in **hexadecimal**.
- **7 separators**.

	IPv4	IPv6
Created	1981	1995 (IETF)
Size	8 × 4 = 32 bits	8 × 16 = 128 bits
Notation	Decimal octets	Hexadecimal groups
Separators	3 (dots)	7

6. TCP/IP Model

- **Transmission Control Protocol / Internet Protocol** model.
- Works on the **transport layer**; known as the **DoD model** (Department of Defense).
- Developed by **ARPA**.
- TCP/IP came first, then the OSI model.



Original TCP/IP model (4 layers) and the modified TCP/IP model (5 layers).

Original (4 layers)	Modified (5 layers)	OSI (7 layers)
Application layer	Application layer	Application, Presentation, Session
Transport layer	Transport layer	Transport
Internet / Network layer	Internet layer	Network
Network Interface	Data link layer + Physical layer	Data link + Physical

The first two columns follow the notes (the Network Interface layer splits into data link + physical); the OSI column is added for comparison.

7. Data Communication

- **Data communication** is the exchange of data between two devices using a communication medium.
- It depends on 4 characteristics: **(a) Accuracy (b) Timeliness (c) Delivery (d) Jitter** (delivering packets to the receiver end).

7.1 Components of data communication

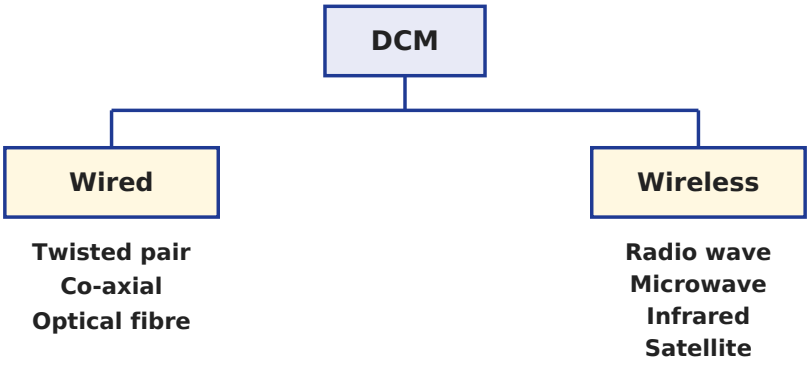
#	Component
1	Sender
2	Receiver
3	Data and message
4	Communication medium
5	Protocols — 3 key elements: (i) **Syntax**, (ii) **Semantics**, (iii) **Timing** (speed of data to send)

7.2 Data transmission modes (responsibility of the physical layer)

Mode	Meaning	Examples
(A) Simplex	Unidirectional — sender to receiver. The sender can only send data but cannot receive; the receiver can only receive but cannot send.	Monitor, keyboard, mouse etc.
(B) Half duplex	Two-way communication is possible, but only **one direction at a time**.	Walkie-talkie
(C) Full duplex	Bidirectional communication **at the same time**.	Mobile, telephone, switch, TCP etc.

7.3 Data communication media

- The way or medium by which data communication takes place. 2 types:



Data communication media (DCM): wired and wireless.

Wired media

(1) Twisted pair

- Two separate **insulated** wires are twisted around each other in a twisted pattern.
- These two wires form a circuit through which data is transmitted.
- When data flows in a wire, a **magnetic field** is created around it; the magnetic fields of the two wires **cancel each other**, which gives protection from crosstalk.
- **Crosstalk** = the noise generated in a nearby wire.

(a) UTP — Unshielded Twisted Pair

- Speed **10 – 1000 Mbps**.
- Uses **AWG** (American Wire Gauge).

- Insulated with **CPE** (cellular polyethylene — as read) or **FEP** (Fluorinated Ethylene Propylene).
- Wires are **colour coded**; each colour shows which application / use the wire is for.
- UTP colour pairs: **Blue-white, Orange-white, Brown-white, Green-white**.

Advantages	Disadvantages
Cheap and easy installation Fastest copper medium	Not useful for long networks Crosstalk is high

(b) STP — Shielded Twisted Pair

- Has an extra **metallic foil shield**; protected from interference coming from inside or outside.
- Both ends must be **grounded**, otherwise the signal leaks.
- Uses the 8-pin **RJ-45** connector.

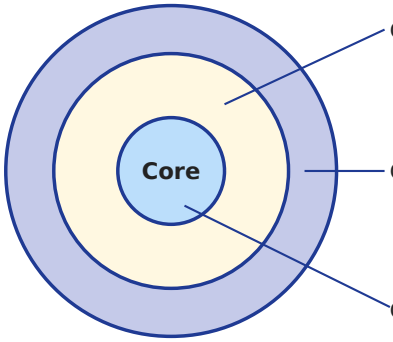
(2) Co-axial cable

- The first type of cable used in networks (also used for Dish TV).
- One **copper wire** (main conductor) covered by a solid electrical **insulator**.
- The insulator is covered by **foil and a metallic mesh** (outer conductor).
- **BNC** (Bayonet Neill–Concelman) connector is used.

	Thin net	Thick net
Standard	IEEE **10Base2**	IEEE standard (written 10base2; 10Base5 is the standard for 500 m)
Max length	185 m	500 m
Speed	10 Mbps	—
Notes	Transmits only a single frequency → called **baseband**; **BNC-T** connector attaches to the NIC	—

(3) OFC / FOC — Optical Fibre Cable

- Contains very thin **glass / fibre filaments** through which data travels as a **light beam**.
- To generate the light beam, an **LED or ILD** (injection laser diode) is used.



Component	Details
(a) Core	Made of glass fibre; passes the light beam; the main conductor — data travels here
(b) Cladding	Protects the core; flexible plastic
(c) Coating	Outer layer; plastic

Cross-section picture added for illustration; the component table is from the notes.

- A layer of **polymer / polyimide** is applied over the cladding.
- It converts coded electronic pulses into **light pulses** and transmits them.
- At the receiver end, the light pulses are converted back into binary data by a **photoreceptor**.
- Data travels using **total internal reflection**.

Mode of OFC	Cost	Bandwidth	Core
(a) Single mode	Costly	High	Small
(b) Multi mode	Cheap	Low	Bigger

Wireless (unguided) media

Medium	Details
(1) Radio waves	Frequency 10 kHz – 1 GHz; bypass obstacles. Types: short range (AM); VHF (Very High Frequency) = FM radio; UHF (Ultra High Frequency) = TV; SHF (Super High Frequency) = 3 – 30 GHz
(2) Microwave	Uses a special type of antenna which sends the signal to the other antenna; works on **line of sight**
(3) Infrared	Secured wireless communication; used in remotes and wireless keyboards; small area (the notes' line on obstacles is hard to read — infrared normally cannot pass through obstacles)
(4) Satellite	A special case of microwave relay; uses a **transponder**; **uplink** (Earth → satellite) and **downlink** (satellite → Earth)

8. Backup and Restoring Data

8.1 (A) RAID configuration

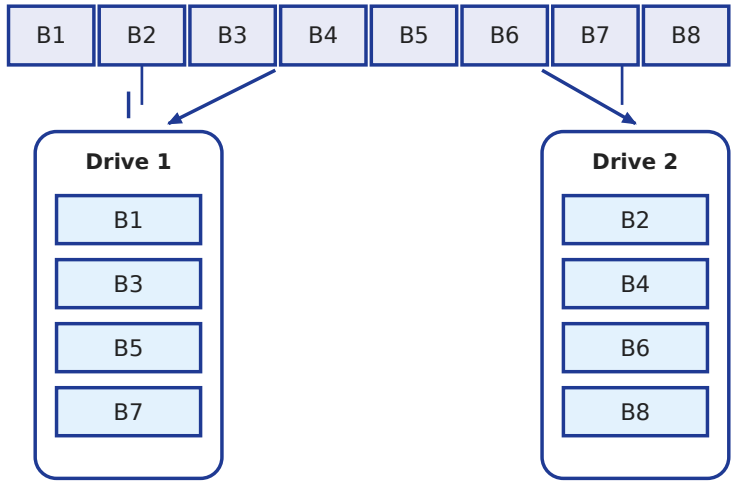
- **RAID** = **Redundant Array of Independent Disks**.
- In a network, data is stored across **multiple disks**.
- Provides **fault tolerance**; better performance and reliability.

Data striping

- Storing parts of data on different hard disks is called **data striping**.
- It is used for **RAID level 0 onwards**.
- Can be of 3 types: **bit level, byte level and block level**.

(1) RAID level 0

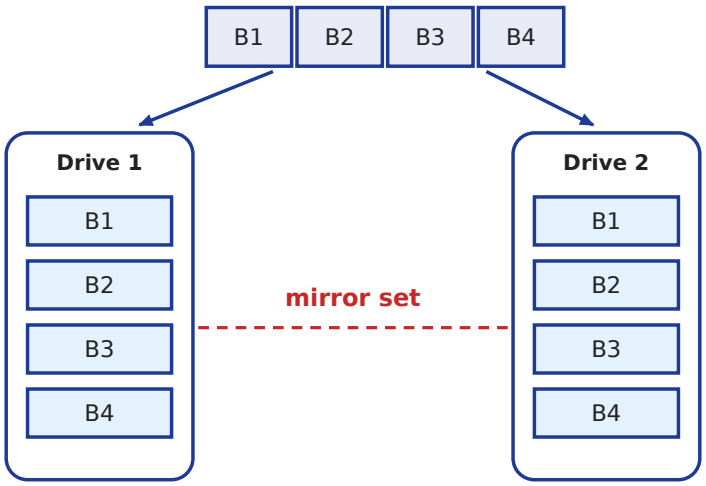
- Uses **block-level striping**; data blocks are stored across multiple disks.
- If any disk's data is lost it **cannot be recovered**.
- Gives the highest / maximum performance (100%) and is the **fastest** RAID level.
- No parity bit is used; **no fault tolerance**.



RAID 0: blocks B1…B8 are striped alternately over Drive 1 and Drive 2.

(2) RAID level 1

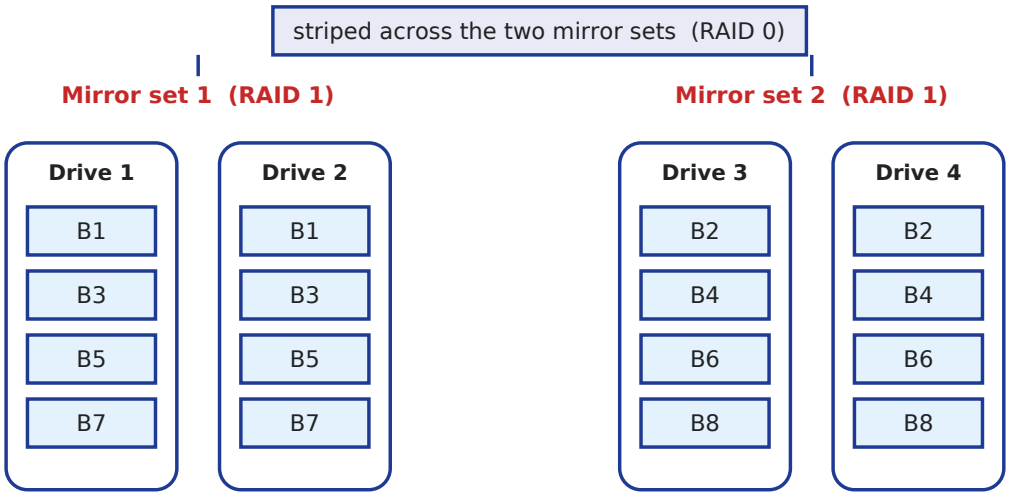
- Also called **shadowing / cloning / mirroring**; block-level striping (as written).
- Minimum **2 disks** are used.
- Performance **50%**.
- No parity used; the storage is a **mirror set**.



RAID 1: the same blocks are written to both drives (mirror set).

(3) RAID level 10 (= RAID 1 + RAID 0)

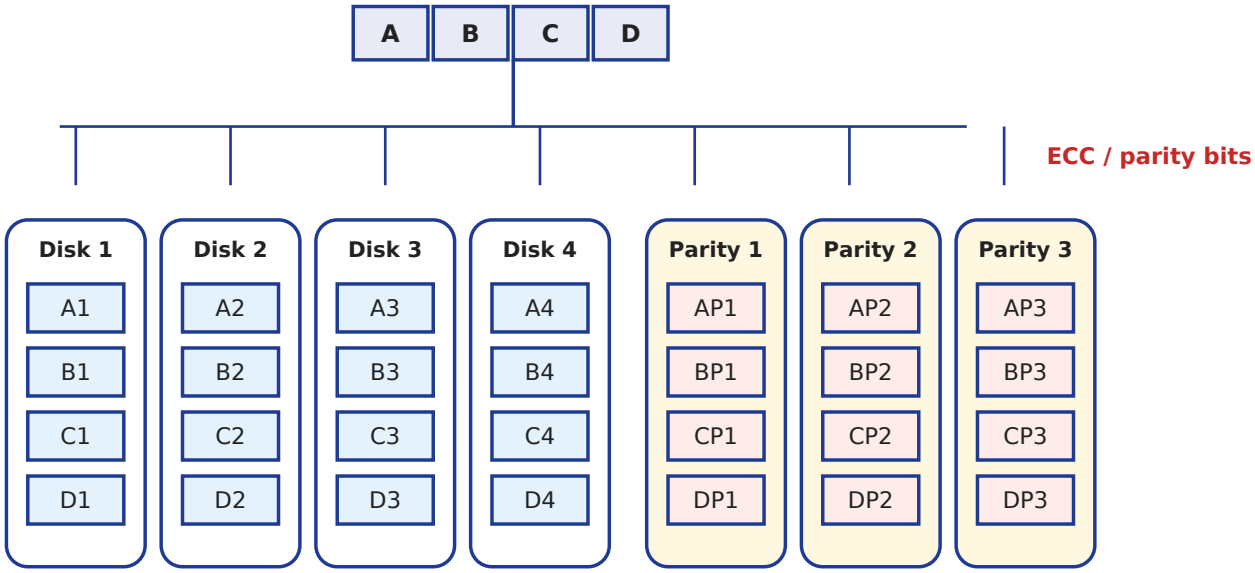
- RAID level 1 + RAID level 0 — **nesting**.
- Block-level striping **and** mirroring.
- Provides performance and reliability.
- No parity used.



RAID 10: two mirrored pairs, striped together.

(4) RAID level 2

- **Bit-level striping** + **ECC** (Error Correcting Code).
- Parity bits are used to **detect and correct** errors.



RAID 2: bits A, B, C, D spread over data disks 1–4, with separate parity disks holding the ECC bits.

(5) RAID level 3

- Uses **byte-level striping + single parity**.
- All parity information is stored in **one dedicated disk**.
- Data can be recovered from the parity bits.
- If the parity disk crashes, data **cannot** be recovered.

(6) RAID level 4

- Uses **block-level striping + parity bits**.
- All parity bits are stored in a **single dedicated disk**.

(7) RAID level 5

- **Block-level striping + single / distributed parity**.

- Data can be recovered if **any one** of the disks crashes, because the parity is **distributed**.

(8) RAID level 6

- **Block-level striping + double distributed parity**.
- Uses the **Reed–Solomon** code.
- More memory needed and costly.

RAID summary

Level	Striping	Parity / copy	Fault tolerance	Notes
0	Block	None	None	Fastest (100%); no recovery
1	Block (as written)	Mirror	Yes	Min 2 disks; performance 50%
10	Block + mirroring	Mirror	Yes	RAID 1 + RAID 0; nesting
2	Bit	ECC parity bits	Yes	Detects & corrects errors
3	Byte	Single parity, dedicated disk	Yes — but not if parity disk fails	
4	Block	Parity, single dedicated disk	Yes	
5	Block	Single, distributed parity	Any one disk may crash	
6	Block	Double distributed parity	Yes	Reed–Solomon; more memory, costly

8.2 (B) Backup

- **Backup** = saving a copy of some data on **another storage**.
- Bringing data from that storage back to the local disk is called **restore**.
- Three types:

Type	What is copied	Points
(i) Full backup	All the data	Needs the minimum time to store / restore (as written); takes the **maximum time to create** the backup; needs more memory and bandwidth
(ii) Differential backup	After the last **full** backup, only the data that has **changed**	—
(iii) Incremental backup	After the last full backup, only the **updated / newly added** data	—

Clarification

The notes say a full backup “needs minimum time to store”; the sentence is most likely about **restoring** (a full backup restores fastest), since the next line says it takes the maximum time to *create*.

Appendix — Clarifications and corrections

The handwritten pages were typed as faithfully as possible. These places were clarified, flagged or marked as added:

Topic	In the handwritten notes	Here
WAN switching	“VLPS (variable length package switching)”	Kept; noted that the usual technique is MPLS
Bluetooth speed	“720 Kbps” (hard to read)	Kept as read
Passive hub	“Can’t work without electricity”	Flagged: passive hubs need no power
Layer-2 frame figure	Trailer at left end, Header at right end	Drawn as in the notes; usual order is Header → Data → Trailer
SONET	“Synchronous Digital Network”	Standard name Synchronous Optical Network noted
Class A range	0–126; public 1.0.0.0 – 127.0.0.0	Kept as written; usable range is usually 1–126
Class B networks	$2^{14} - 2 = 16384$	Kept; arithmetic gives 16,382
Class E prefix	5 bits 1,1,1,1,0 with range 240–255	Standard prefix is 1111; flagged
Thick net	“10base2”, 500 m	Flagged: 500 m belongs to 10Base5
Session protocol RPC	“RP (Remote Control)”	Remote Procedure Call
Full backup	“Needs minimum time to store”	Probably “restore”; flagged
Added for revision	—	Comparison tables, summary tables, OFC cross-section picture, TCP/IP vs OSI column, RAID summary table